Abstract

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This document is a **template**, not an empirical study. It demonstrates the registered-report workflow end to end: locking a preregistration, validating its completeness, executing the registered analysis plan against deterministic demonstration data, and reporting confirmatory and exploratory claims through an explicit deviation ledger. Every quantity reported here is produced by the tested code in `src/registered_report/` and regenerated by `scripts/generate_figures.py`; none is hand-entered or illustrative.

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The demonstration binds a single confirmatory hypothesis (H1) to one registered outcome (`primary_score`) analysed by a two-sided label-permutation test. Run on a seeded two-group dataset (`seed = 20260709`, `n = 24` per group), the registered test yields an observed mean difference of `1.003` with a two-sided permutation p-value of `0.0005` (`0` of `2000` shuffles at least as extreme), significant at the preregistered `alpha = 0.05`. A deliberately introduced secondary endpoint and an alternative model are carried only as **documented deviations**, keeping the confirmatory claim boundary intact, and a preregistered sensitivity analysis on the primary outcome is validated and reported alongside the confirmatory result. The purpose is to give forks a working, auditable skeleton in which planned analyses, frozen hypotheses, deviations, and post-run claims are

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separable and machine-checkable.